

THE HEAVY ION STORAGE AND COOLER RING PROJECT ESR AT GSI

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The Experimental Storage Ring ESR [1] is designed for accumulation, storage and cooling of heavy ion beams up to uranium with energies up to 500 MeV/u either directly out of the heavy ion synchrotron SIS [2] or via a stripper target, where electron-free or electron-poor ions are produced. The intense, cold beams may be used for a large variety of experiments, preferably in-ring experiments making use of the special features of the storage ring. Due to circulation frequencies of up to 2 MHz high luminosities will be attained using thin internal gas jet or electron target. Energies can be varied over a wide range from 500 MeV/u down to 3 MeV/u. The SIS/ESR-facility will also be capable of delivering beams of unstable nuclei produced by fragmentation or fission of heavy projectiles. After preselection in a large acceptance Z/A -separator the secondary beams can be prepared for internal or external experiments by stochastic precooling, intensity accumulation and final electron cooling.

1. Introduction

At GSI heavy ions up to uranium have been accelerated in the Unilac [3] since 1975. The maximum specific projectile energy of 10 MeV/u was upgraded to 20 MeV/u in 1982. Plans to extend the accelerator facility by the addition of the synchrotron SIS ("Schwer-Ionen Synchrotron") and the storage and cooler ring ESR ("Experimentier-Speicher-Ring") were approved in 1985 [2]. A plan view of the SIS/ESR-facility is given in fig. 1.

The SIS is a rather fast cycled, free programmable, separated function machine with maximum magnetic bending power of 18 Tm. The specific energy of ions can be varied from 30 MeV/u to 2 GeV/u for Ne^{10+} and to nearly 1 GeV/u for U^{92+} . The Unilac delivering beam pulses of typically 5 ms length with 50 Hz will be modified for time sharing between low energy experiments and injection into SIS. Multiturn injection into the synchrotron requires up to 5 pulses of 0.5 ms length per second with very high pulse currents, e.g. roughly 1 pA for uranium. In order to meet these intensity requirements also for ions with mass numbers $A \geq 40$, the Unilac will be equipped with an RFQ-injector. If the SIS is filled up to the space charge limit ion numbers per SIS-cycle of $2 \times 10^{11} \text{ Ne}^{10+}$ and $3 \times 10^{10} \text{ U}^{72+}$ are attained. The corresponding mean ion rates after acceleration in the SIS at 30 MeV/u are $10^{12}/\text{s}$ and $1.5 \times 10^{11}/\text{s}$, respectively. At maximum energy these rates are reduced by roughly a factor 5 because of reduced repetition frequency.

The SIS will supply experiments with projectiles directly or via ESR, where beams – possibly after being stripped – can be intermediately stored and cooled. The prime goals of the SIS/ESR facility are to provide the following:

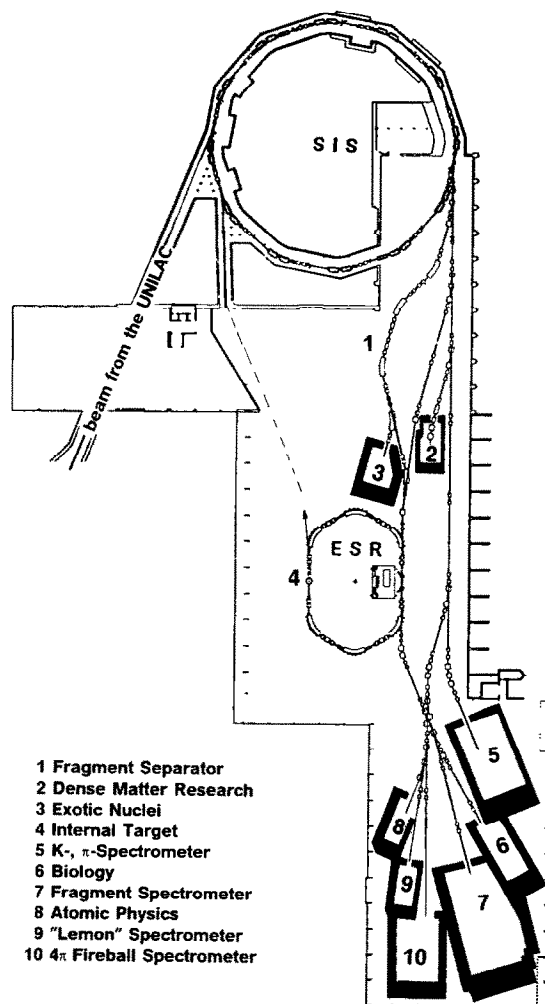


Fig. 1. Plan view of the SIS/ESR-project at GSI.

- Completely stripped heavy ion beams up to U^{92+} with phase space densities as high as possible using various cooling techniques in the ESR. The cooled beams can be used for in-ring experiments or be slowly extracted for external use. The duty time for external experiments can be increased to nearly 90% by using the ESR as stretcher ring for SIS.
- Radioactive heavy ion beams by accumulation and cooling projectile fragmentation or fission products at energies of about 500 MeV/u.
- Facilities for experiments in the ESR using internal gas or electron targets or for studies of ion interaction with laser light.
- Storage of two cooled, co-circulating beams in the ESR. By a special mode of operation, the beams can be crossed and collisions of exotic partners such as for example $U^{92+} \rightarrow U^{92+}$ at well defined energies near the Coulomb barrier can be studied [4].
- Completely stripped, cooled beams of heaviest ions for further acceleration or deceleration in the SIS. By this way, the maximum energy for uranium ions can be raised by 35% to 1.35 GeV/u.

The long term goal is to use the Unilac/SIS/ESR-facility as injection chain for intersecting, superconducting storage rings for colliding beam experiments with $U \rightarrow U$ at relativistic center-of-mass energies up to 20 GeV/u. A recently considered small collider for c.m. energy of about 1 GeV/u may be a reasonable step into that direction. The hope is that beam cooling could help to keep the costs of the collider projects as low as possible.

2. Design features of ESR

Main features of the ESR design are the maximum bending power of 10 Tm, large transverse and momentum acceptances, stochastic precooling and electron cooling of beams, and a special magnet structure with two 10 m long, magnet-free sections leaving sufficient space for electron cooling and equipment for in-ring experiments. The technical layout of the ESR is shown in fig. 2; the major parameters are listed in table 1. The ring geometry is a stretched hexagon determined by 6 bend magnets and two long straight sections. The superimposed focusing structure, a mixture of 4 quadrupole doublets and 4 triplets, turned out to be the best solution to adapt ring optics to the different tasks of the ESR: high performance for beam accumulation, storage and cooling, reasonable technical solutions for beam injection, fast ejection to SIS and slow extraction to the target hall and, not least, suitable conditions for different kinds of in-ring experiments. Ion optical modes with different envelope and dispersion functions at a fixed working point (Q_h , Q_v) allow either optimal ring acceptances or optimal beam matching to electron cooling

and internal target or special modes as multicharge operation or the rather exotic crossed beam mode (mode III in table 1). Even dynamical transitions between ion optical modes are possible if required.

The circumference C of the ESR represents the result of carefully balancing the advantages of small ring size against the demand for easy beam transfer and comfortable space for special equipment for beam cooling and in-ring experiments. Electron cooling rates as well as luminosities of experiments using an internal target are $\propto C^{-1}$. The luminosity attainable in the crossing volume of two co-circulating beams is even $\propto C^{-2}$. The cost estimate due to large transverse and momentum acceptances of the ring gave additional arguments for a ring as small as consistent with the design goals. Hence, the ESR size was finally fixed at 108.36 m, exactly one half of the SIS-circumference. As the same ratio (2:4) was also chosen for the harmonic numbers of the rf-accelerating cavities, the rf-range is identical for both rings.

Fast beam transfer from SIS into ESR has to be realized by two steps. In each step, two of the four SIS-bunches are captured into rf-buckets and deposited in the ESR by means of the rf-stacking technique. Repeated application of this procedure allows accumulated beam currents up to space charge limits. The attainable beam intensities are strongly correlated with the desired ion energy and with the requirements for transverse emittances, momentum spread, and eventually time structure of the ion beam.

Two cavities with maximum voltage amplitudes of 5 kV and tunable in the rf range 0.85–5 MHz are provided for rf-stacking, beam bunching and beam energy variation. The beam, in general highly ionized ions stored and cooled at high energy, can be decelerated down to 11.5 MeV/u without changing the harmonic number $h = 2$. Transition from $h = 2$ to $h = 4$ at medium energy near 100 MeV/u is necessary for further deceleration down to 3 MeV/u, the minimum energy of the ring. The maximum field ramp of $\dot{B} = \pm 1$ T/s requires an effective voltage per turn $U_{\text{eff}} = C\dot{B}\rho = \pm 677$ V, where ρ is the orbit radius in the bending magnets. Deceleration from energies listed in table 1 down to 10 MeV/u needs about 1.5 s. Small energy changes of a coasting beam can be done by changing slowly the energy of cooling electrons.

The most severe source of beam losses in the ESR are charge changing collisions between highly charged ions and residual gas atoms. The most critical process, electron capture at low energies, requires an ultrahigh vacuum system of 10^{-11} mbar pressure. The estimated lifetime of a U^{92+} -beam at 20 MeV/u is about 100 s. One important technical complication due to this low pressure is the necessity of in situ baking of the system at 300°C, including all equipment for beam cooling and in-ring experiments.

I. ATOMIC PHYSICS / RELATED PHENOMENA

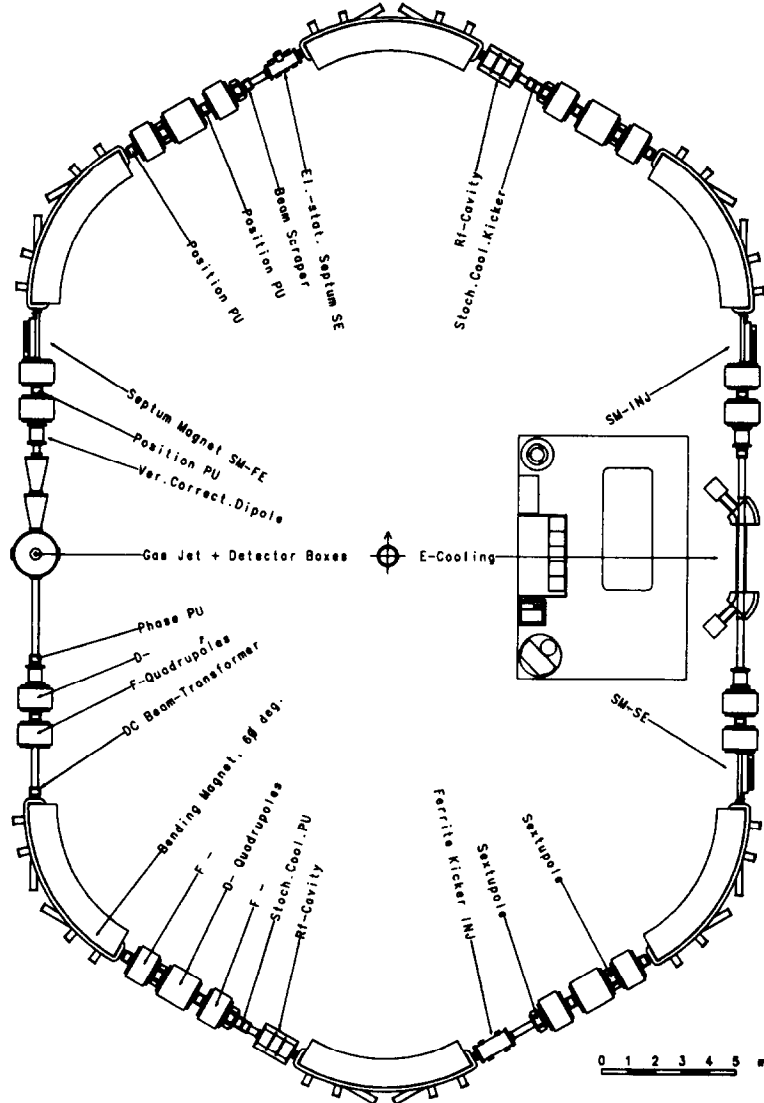


Fig. 2. Layout of the Experimental Storage Ring ESR. The main parameters are listed in table 1. Beam injection and extraction channels are shown in fig. 1.

3. Ion beam cooling

Stochastic cooling [5] (SC) plays in the ESR the role of a precooling stage for secondary beams. Hot beams of projectile fragments will be produced in a rather thick target and selected by a magnetic separator (see fig. 1). For the cooling of 10^8 medium heavy fragment ions at 500 MeV/u precooling will start at transverse emittances $\epsilon = 20 \pi \mu\text{m}$ and momentum spread $\delta p/p = \pm 5 \times 10^{-3}$. Electron cooling [6] (EC) is applied at $\epsilon = 2 \pi \mu\text{m}$ and $\delta p/p = \pm 5 \times 10^{-4}$. For final values $\epsilon = 0.1 \pi \mu\text{m}$ and $\delta p/p = \pm 1 \times 10^{-4}$ by consecutive application of SC and EC a time of about 1 s will be

necessary. This may be considered also as a lower limit for lifetimes of the radioactive nuclei to be stored, cooled and experimented with in the ESR. Final beam temperatures will, of course, depend on number, energy and charge-to-mass ratio of stored ions. These parameters determine space charge limit (SCL), microwave instabilities (MWI) or intrabeam scattering (IBS) growth rates [7]. Vice versa, the maximum beam intensity is determined mainly by the same limitations. Ion numbers given in table 1 are SCL for incoherent betatron tune depression of $\Delta Q_{h,v} = 0.05$, maximum energies and coasting beams with $\epsilon = 1 \pi \mu\text{m}$ and $\delta p/p \approx \pm 2.5 \times 10^{-4}$.

Table 1
Major ESR-parameters

<i>General</i>			
Specific energy	proton	30–2200	MeV
(ion numbers, see section 3)	$^{20}\text{Ne}^{10+}$ (1.3×10^{11})	3.0–834	MeV/u
	$^{20}\text{Ar}^{18+}$ (6.2×10^{10})	3.0–709	MeV/u
	$^{84}\text{Kr}^{36+}$ (2.9×10^{10})	3.0–656	MeV/u
	$^{129}\text{Xe}^{54+}$ (1.7×10^{10})	3.0–609	MeV/u
	$^{238}\text{U}^{92+}$ (9.3×10^9)	3.0–556	MeV/u
Magnetic rigidity		0.5–10	Tm
Electrostatic rigidity		10–2550	MV
Ring circumference		108.36	m
Mean radius		17.246	m
<i>Ion optics</i>			
Optical mode		I/II/III	
Betatron tune	Q_h (horiz.)	2.2/2.2/2.2	
	Q_v (vert.)	2.4/2.4/2.4	
Transition point	γ_t	2.66/2.57/2.85	
Acceptances	A_h ($\delta p/p = 0$)	411/383/554	$\times 10^{-6} \pi \text{ m}$
	A_v	163/164/170	$\times 10^{-6} \pi \text{ m}$
	$\Delta p/p_{\max}$ ($\epsilon_h = 0$)	$\pm 36/\pm 20/\pm 17$	$\times 10^{-3}$
<i>Main magnets</i>			
Bending magnet	number $\times$ angle	6×60	deg
	field	0.08–1.6	T
	bending radius	6.25	m
	used aperture ($h \times v$)	220×70	mm^2
	field ramp	± 1	T s^{-1}
Quadrupoles	number (short/long)	16/4	
	field gradient	0.31–6.1	T m^{-1}
	effective length	0.75/117	m
	used aperture ($h \times v$)	300×150	mm^2
<i>Rf-cavities</i>			
Number $\times$ voltage		2×5	kV
Frequency range		0.85–5	MHz
Harmonic number		2 or 4	
<i>Beam diagnosis</i>			
Position pick up	(number)	12	
Phase pick up		2	
Schottky pick up		3	
Beam dc transformer		1	
<i>Injection / accumulation / extraction</i>			
Inflection	magnet type	copper septum	
	angle	80	mrاد
Ferrite kickers	rise time	90	ns
	total angle	4	mrاد
Accumulation	method	rf-stacking	
	time per step	≥ 20	ms
Slow extraction	method	3rd int. res.	
	beam septum	wires	
	angle	4	mrاد
	field	80	kV/cm
<i>UHV-system</i>			
Working pressure		10^{-11}	mbar
Baking temperature		300	$^{\circ}\text{C}$

Table 1 (continued)

<i>Ion beam cooling</i>			
Stochastic cooling	ion energy	500	MeV/u
	$\delta p/p$ range	1–0.1	%
	$\epsilon_{h,v}$ range	20–2	$10^{-6} \pi \text{ m}$
	time for U^{92+}	≈ 1	s
Electron cooling	ion energy	30–560	MeV/u
	$\delta p/p$ range	0.1–0.001	%
	$\epsilon_{h,v}$ range	2–0.05	$10^{-6} \pi \text{ m}$
	time for U^{92+}	≈ 0.2	s
<i>Internal gas jet target</i>			
Gas jet	diameter	3–5	mm
	density	$10^{14} - 10^{17}$	mol cm^{-3}
	material	$\text{H}_2, \text{N}_2, \dots$	
Target thickness	(H_2)	0.3–300	ng cm^{-2}

3.1. Stochastic precooling

The technical design of the SC components for ESR is in a very preliminary stage. Though incorporated into a more or less frozen ESR lattice SC turned out to be well feasible for a tolerable technical expenditure. One of the reasons for this is that, at the planned operation point $\gamma = 1.6$, the ratio η between frequency spread $\delta f/f$ and momentum spread $\delta p/p$ of ions has the relatively large value of 0.3. This fast wanted mixing enables high gains between SC pickups and kickers to be applied. In addition, highly charged heavy ions have, compared to protons, the inherent advantage of higher Schottky signal which allows either cooling at higher beam intensity or cooling with higher speed to lower equilibrium beam temperatures. The price to be paid is that, under given conditions and for the same cooling rate, the power requirements increase $\propto A^2/Z^2$. For the cooling of 10^8 uranium-like nuclear fragments in either of the transverse phase planes from $20 \pi \mu\text{m}$ to $2 \pi \mu\text{m}$ within 1 s between 1 and 2 kW power in the frequency band 0.8–1.6 GHz would be needed, depending on the

length of the kicker [8]. Momentum cooling can be done simultaneously with comparable expense of power. The time delay due to intrinsic amplifier capacitances has to be shorter than 90 ns if pickups and kickers are arranged on opposite positions in the ring (see fig. 2). Our hope is focused on the quite advanced development of a solid state amplifier for the ACOL ring at CERN. 30 dB gain, 48 ns time delay and 100 W power in the frequency band 0.8–1.6 GHz are the parameters for a three stage module [9].

3.2. Electron cooling

A few selected technical parameters of the EC section [10] are listed in table 2, the technical layout can be seen in fig. 3. The effective length is 2.5 m, but with all the necessary equipment, included magnetic steerers, compensation solenoids etc., 9.5 m are needed. The maximum electron beam energy of 310 keV corresponds to the maximum specific energy of U^{92+} -ions in the ESR. Electron currents of up to 10 A at the highest energy are planned for beam diameters between 3 and 5 cm. A variable longitudinal magnetic guiding field of up to 0.25 T will conserve the beam diameter. Space charge forces cause collective beam rotation and parabolic velocity increase with increasing distance from the beam axis. Our simulation of EC, based on the theoretical model for the magnetized electron plasma [11], includes these effects as well as special properties of the ESR lattice. Optimal cooling rates are obtained with certain correlations between ion energy, magnetic solenoid field and density of electron current [12]. Fig. 4 illustrates the time dependence of beam parameters $\epsilon_{h,v}$ and $\delta p/p$ for U^{92+} -ions during the EC-process at 477 MeV/u.

EC will extend storage time or beam lifetime during experiments using the internal gas target. Working as compensation for coherent space charge forces driving

Table 2
The main parameters of the electron cooling section

Operating range	30–560	MeV/u
Electron energy	16.5–310	keV
Electron beam	current	5–10 A
	current density	0.5–1.0 A cm^{-2}
Cathode	diameter	30–50 mm
	temperature	1100 °C
Guiding field	strength	0.01–0.25 T
	power loss	≤ 300 kW
Free aperture	diameter	250 mm
Length	effective	2.5 m
	installation	6.0 m

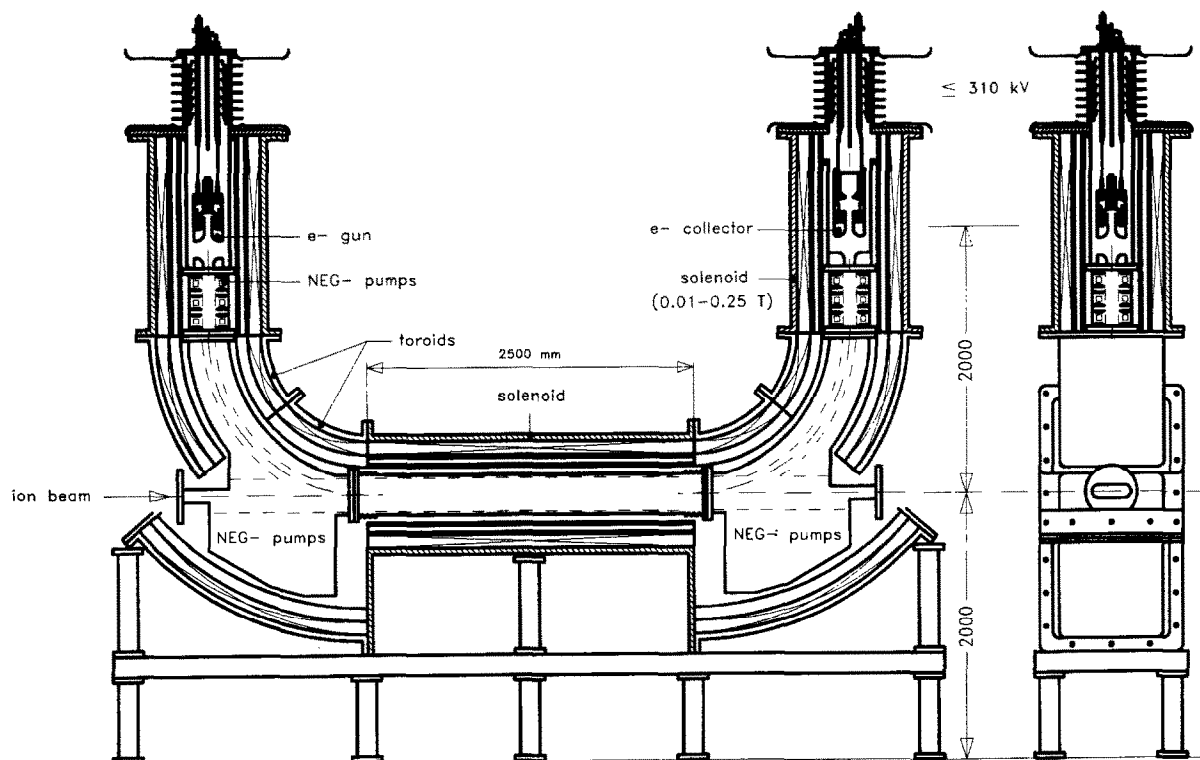


Fig. 3. Schematic drawing of the electron cooling section [10] of ESR. Technical parameters are listed in table 2.

e.g. MWI or beam heating due to IBS and scattering by target nuclei, it will stretch considerably the duty time and efficiency. It was demonstrated by computer simulation for the ESR-lattice [13] that IBS can dilute the beam within a few seconds, but also that this dilution can be stopped by means of EC at rather narrow equilibrium phase space distributions.

An important limitation of the beam life due to EC are radiative electron capture (REC) processes in the cooling section with rates increasing $\propto Z^2$. The life time estimate for a cooled U^{92+} -beam is 20 s. As beam injection and cooling will require less than 1 s, one may expect duty factors of up to 90% for in-ring experiments with permanent electron cooling.

4. Outlook on the experimental programme

Though there are many attractive applications for very cold heavy, possibly radioactive, ion beams outside the ESR, we restrict the discussion to in-ring experiments, i.e. the use of the storage ring as an experimental facility. The experimental programme [14] is mainly keyed to the special features of the ESR and to its connection to the heavy ion accelerator facility Unilac/SIS. The large variety of experimental questions preferably studied in the ESR may be deduced from the

following – necessarily preliminary and incomplete – overview.

4.1. Nuclear physics

Storage ring mass spectrometry: Precise mass determination of rare nuclei, such as ^{205}Pb or exotic projectile fragments requires extremely cold beams ($\delta p/p \leq 10^{-5}$) as well as highly stabilized and reproducible bending and focusing fields. Once mean ion velocity spread are in equilibrium with electron cooling, absolute masses or mass differences can be derived from measurements of circulating frequency distribution (Schottky-scan [15]). Under suitable conditions, i.e. high ionic charge and sufficient beam intensity, we hope to achieve a relative precision $\delta A/A$ between 10^{-5} and 10^{-6} .

Measurements of nuclear moments: The strong hyperfine interaction of heavy few-electron ions can be used to study electromagnetic moments of highly excited short lived nuclear states or radioactive nuclei.

Investigation of β -decay: Some nuclides will lose β -stability if the inner electrons are removed. As decay rates are generally rather low, observation of these processes will require long storage time in the order of minutes. Decay rates will be determined by measuring the fraction of daughter nuclei as a function of storage time. Of cosmological interest is the decay $^{187}_{75}\text{Re}^{75+} \rightarrow ^{187}_{76}\text{Os}^{75+}$.

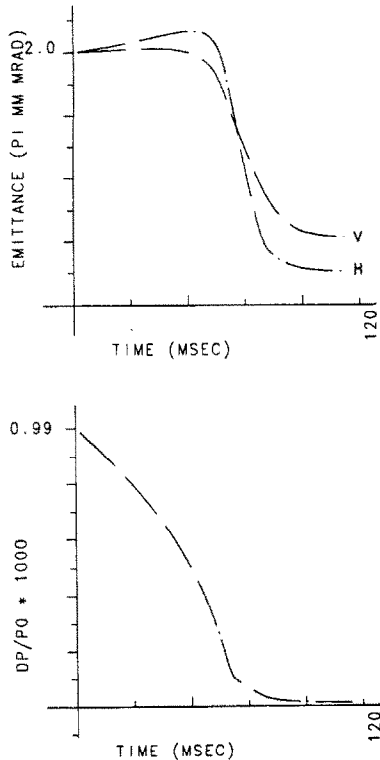


Fig. 4. Theoretically [11] calculated emittances and momentum spread ($DP/PO = \delta p/p$) of a U^{92+} -beam at 477 MeV/u during electron cooling in the ESR [12]. The cooling process seems to be stopped at ion beam temperatures where the theoretical model ceases to be applicable. Electron beam parameters: current 10 A, diameter 5 cm, solenoid field 0.2 T, e^- -temperatures 0.3 eV in transverse and 0 eV in longitudinal phase planes.

Nuclear reaction studies: Direct reactions between circulating heavy, possibly radioactive, ions at 200–400 MeV/u and light target nuclei as H, D etc. may be studied by using the light recoils as sensitive probes. The special attraction of this method of *inverse kinematics* is the access to a reaction spectroscopy of rare or short lived nuclei for which targets are not available.

4.2. Atomic physics

Ion–electron interactions: Once the ion beam is already cooled the cooling electrons may be used as an excellent electron target by means of fast changes of the electron energy. Cross sections for REC or di-electronic recombination [16] can be measured as functions of the electron energy this way. Even laser induced REC into highly excited states may be observed [17]. If the REC cross section functions are already known measurements might be applied for ion beam temperature diagnosis.

Ion–ion collisions: Collisions between two totally stripped or few electron heavy partners at energies up to the Coulomb barrier can be studied by crossing two beams [4]. The beams will co-circulate in the ESR at several 100 MeV/u on separate closed orbits due to slightly different momenta. By means of a special ion optical mode and additional quadrupoles the beams will cross each other at two positions. The luminosity, estimated from space charge and microwave stability limits, for the most interesting system $U^{92+} \rightarrow U^{92+}$ is $7 \times 10^{23} \text{ cm}^{-2} \text{ s}^{-1}$. It increases very strongly with lower charge state $\propto A^2 q^{-4}$. Charge exchange, δ -electron or quasi-molecular X-ray emission may be investigated within a large fraction of solid angle. Even a very clean observation of spontaneous electron-positron pair creation in the high Coulomb field of two naked uranium ions seems to be possible though the preliminary event rate estimate of 10^{-3} s^{-1} is very low [18].

Ion–atom collisions: Various electron capture and loss processes in a wide range of ions and energies can be studied making use of the internal gas jet target. Electron capture in collisions between totally stripped high-Z ions and low-Z target atoms define very clean conditions for an interesting and – if performed at low energy – high precision atomic spectroscopy. QED effects can be investigated by measuring higher order effects in the Lamb-shift which are at the 10^{-6} level in neutral hydrogen but dominate in H-like uranium.

Ion–photon interactions: Resonant excitation of ions by means of collinear laser radiation may be possible due to the high phase space density of cooled ion beams and by applying high laser pulse power with amplification in a resonant mirror system. Resonant 1s hyperfine splitting and the resulting bound-state g -factor in one-electron heavy ions might be determined this way [19].

4.3. Plasma and accelerator physics

Production of hot and dense plasma: Stopping of fast heavy ion beams in gaseous or solid matter causes extremely high ionisation and excitation density, especially if the beams are accumulated, cooled and bunched to peak intensities and phase space densities as high as possible. Production and investigation of inertially compressed, hot plasma spots in solids will open a wide field for astrophysical studies and will deliver more information about feasibility of heavy ion induced fusion [20]. Heating of a gaseous target by cooled ion beams at energies of some 100 MeV/u could be probably used to pump soft-X-ray gas lasers [21].

Heavy ion storage and cooling: Storage and cooling of heavy ions are new fields in accelerator physics which will, of course, be subject of experimental studies at the ESR. Topics of interest are, for instance, investigations of MWI and IBS, tests of new cooling techniques or development of new methods for beam diagnosis.

5. Project status and time schedule

The current project status is as following: The contract for the concrete work was awarded in spring, but forest clearing and site preparation had to wait until October 1st, 1986. The tunnel for SIS and the power supply building will be ready for installation at the beginning of 1988, the hall complex for ESR and experiments 9 month later. Main magnets with power supplies are ordered. The major part of the components for beam injection and extraction are designed or under construction. Activities on other ring components are already included in the work for SIS.

The start of ESR operation is scheduled for January 1990, one year after the first beam injection into SIS. The electron cooling section and the internal gas jet target should be available at this date whereas stochastic cooling will be operated with a preliminary momentum cooling stage. Presently, the first generation of ESR experiments is being defined and GSI is going to call for proposals.

This paper necessarily reports mostly on the cooperative work of numerous colleagues from GSI and many other institutes. I am grateful to all who have contributed to the design or are contributing to the realization of the ESR facility.

References

- [1] B. Franzke, H. Eickhoff, B. Franzak and B. Langenbeck, internal GSI-Report, GSI-SIS-INT/84-5 (1984) and B. Franzke, IEEE Trans. Nucl. Sci. NS-32 (5) (1985) 3297.
- [2] K. Blasche, D. Böhne, B. Franzke and H. Prange, IEEE Trans. Nucl. Sci. NS-3 (5) (1985) 2657, and ref. [14].
- [3] J. Glatz, Linear Accelerator Conference, Stanford (1986).
- [4] B. Franzke and Ch. Schmelzer, IEEE Trans. Nucl. Sci. NS-32 (5) (1985) 2678.
- [5] S. Van der Meer, CERN-ISR-PO/72-31 (1972).
- [6] Ya. Derbenev and A.N. Skrinsky, Part. Accel. 8 (1978) 235.
- [7] H. Herr, Workshop on Nuclear Physics with Stored Cooled Beams, Bloomington, IN, USA (1984).
- [8] A.G. Ruggiero, Design of the Stochastic Cooling for the ESR at GSI (1986) unpublished.
- [9] G. Garron, F. Caspers and L. Thorndahl, Cern 85-01 (1985).
- [10] N. Angert, H. Beyer, B. Franzke, B. Langenbeck, D. Liesen, F. Nolden, H. Schulte, P. Spädtke and B. Wolf, GSI-Scientific Report 1985, GSI 86-1 (1986) p. 370.
- [11] A.H. Sørensen and E. Bonderup; Nucl. Instr. and Meth. 215 (1983) 27.
- [12] F. Nolden, Internal GSI-Report GSI-ESR/86-02 (1986).
- [13] Z. Meyer-Prüssner et al., GSI-Scientific Report 1985, GSI 86-1 (1986) p. 388.
- [14] P. Kienle, The SIS/ESR Project of GSI, GSI-Report, GSI 85-16 (1985).
- [15] J. Borer, P. Bramham, H.G. Hereward, K. Hübner, W. Schnell and L. Thorndahl, Proc. IXth Int. Conf. on High Energy Accelerators, Stanford (1974) p. 53.
- [16] H. Hahn, Int. Seminar on Dynamic Processes of Highly Charged Ions, Susono, Japan (Aug. 21–24, 1986).
- [17] R. Neumann et al., Z. Phys. A313 (1983) 253.
- [18] H. Backe, Workshop on the Physics with Heavy Ion Cooler Rings, Heidelberg (1984).
- [19] T. Kühl and G. Huber; Arbeitstreffen über Experimente am geplanten Experimentier-Speicherring der GSI, Rauschholzhausen, FRG (1984).
- [20] K. Beckert et al., GSI-Report, GSI 83-10 (1983).
- [21] A. Ulrich, H. Bohn, P. Kienle and G.J. Perlow, Appl. Phys. Lett. 42 (1983) 782.